

fuse_modules#

https://pytorch.org/docs/stable/generated/torch.ao.quantization.fuse_modules

Description:

Fuse a list of modules into a single module. Fuses only the following sequence of modules: conv, bn conv, bn, relu conv, relu linear, relu bn, relu All other sequences are left unchanged. For these sequences, replaces the first item in the list with the fused module, replacing the rest of the modules with identity. model – Model containing the modules to be fused modules_to_fuse – list of list of module names to fuse. Can also be a list of strings if there is only a single list of modules to fuse. inplace – bool specifying if fusion happens in place on the model, by default a new model is returned

Function Signature

```
class torch.ao.quantization.fuse_modules.fuse_modules(model,
modules_to_fuse, inplace=False, fuser_func=<function
fuse_known_modules>, fuse_custom_config_dict=None)[source]#
```

Parameters

Returns

Returns:

model with fused modules. A new copy is created if inplace=True.

Code Examples

```
# Example of fuse_custom_config_dict
fuse_custom_config_dict = {
    # Additional fuser_method mapping
    "additional_fuser_method_mapping": {
        (torch.nn.Conv2d, torch.nn.BatchNorm2d): fuse_conv_bn
    },
}
```

```
>>> m = M().eval()
>>> # m is a module containing the sub-modules below
>>> modules_to_fuse = [ ['conv1', 'bn1', 'relu1'],
    ['submodule.conv', 'submodule.relu']]
>>> fused_m = torch.ao.quantization.fuse_modules(m,
modules_to_fuse)
>>> output = fused_m(input)

>>> m = M().eval()
>>> # Alternately provide a single list of modules to fuse
>>> modules_to_fuse = ['conv1', 'bn1', 'relu1']
>>> fused_m = torch.ao.quantization.fuse_modules(m,
modules_to_fuse)
>>> output = fused_m(input)
```

Detailed Documentation

Documentation

Fuse a list of modules into a single module.

Fuses only the following sequence of modules: conv, bn conv, bn, relu conv, relu linear, relu bn, relu. All other sequences are left unchanged. For these sequences, replaces the first item in the list with the fused module, replacing the rest of the modules with identity.

`model` – Model containing the modules to be fused

`modules_to_fuse` – list of list of module names to fuse. Can also be a list of strings if there is only a single list of modules to fuse.

`inplace` – bool specifying if fusion happens in place on the model, by default a new model is returned

`fuser_func` – Function that takes in a list of modules and outputs a list of fused modules of the same length. For example, `fuser_func([convModule, BNModule])` returns the list `[ConvBNModule, nn.Identity()]`. Defaults to `torch.ao.quantization.fuse_known_modules`

`fuse_custom_config_dict` – custom configuration for fusion

model with fused modules. A new copy is created if `inplace=True`.